

# Zein Shehabeddine

MSC. COMPUTER SCIENCE

Beirut, Lebanon

☎ (+961) 79-161025 | ✉ zein@zeinshehab.com | 🏠 zeinshehab.com | 📷 zeinshehab | 🌐 zeinshehabeddine

## Summary

I am a computer scientist interested in applied machine learning, software engineering, and bioinformatics. My work spans clinical outcome prediction, genetic variant classification, sign language technologies, and large-scale bacterial genomics.

## Education

### American University of Beirut (AUB)

MASTER OF SCIENCE (MS) IN COMPUTER SCIENCE [GPA: 4.0]

Beirut, Lebanon

2024 – 2026

### American University of Beirut (AUB)

BACHELOR OF SCIENCE (BS) IN COMPUTER SCIENCE [GPA: 3.7, Major GPA: 4.0]

Beirut, Lebanon

2021 – 2024

- Education Above All Scholarship

## Experience

### American University of Beirut (AUB)

GRADUATE RESEARCH ASSISTANT

Beirut, Lebanon

2025 – 2026

- Conducted research in applied machine learning and bioinformatics across clinical outcome prediction, genetic variant classification, autism speech analysis, and large-scale bacterial operon prediction.
- Developed machine learning and data analysis pipelines for clinical, genomic, speech, and bacterial genome datasets.

### American University of Beirut (AUB)

GRADUATE ASSISTANT

Beirut, Lebanon

2024 – 2025

- Taught computer science labs and recitations covering data structures and algorithms, Java programming, cybersecurity, and Microsoft Excel/Access.
- Developed lab exercises and assessment questions, prepared instructional materials, and graded lab assignments and examinations.

## Selected Projects

### OperonAtlas — operonatlas.org.

Built and deployed the largest-scale database of computationally predicted bacterial operons, spanning 21,000+ bacterial genomes and nearly 90 million genes. Developed the data and inference pipelines and a production web platform for searching, visualizing, comparing, and downloading predicted operons using Cloudflare Workers, D1, R2, and Pages.

### Labeeb: Lebanese Sign Language Translator.

Built the first Lebanese Sign Language translation app end-to-end, from data collection and TensorFlow/MediaPipe models to a Flask REST API and Flutter mobile frontend. Led to a preprint on lightweight Arabic Sign Language recognition using landmark-based representations.

### Connect Four AI.

Led a team of four to design and implement a high-performance Connect 4 AI in C using negamax with iterative deepening and null-window search optimizations. The bot won an internal department competition against other student bots and outperformed public online bots as well.

## Skills

### Programming

Python, C, Java, JavaScript, SQL, Bash, HTML/CSS

### Machine Learning

PyTorch, TensorFlow, scikit-learn, Pandas, NumPy, OpenCV, Deep Learning, Statistical Analysis

### Systems

Linux, Docker, HPC

### Tools & Platforms

Git, CI/CD, REST APIs, SQLite, Cloudflare, Oracle Cloud

# Publications

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**A Framework for BRCA1/2 Variant Interpretation in Data-Limited Populations Integrating Global Supervised Learning and Population-Specific Anomaly Detection**

Manuscript under review

R. ASSAF, Z. SHEHABEDDINE, H. SAAD, N. ASSAF2026

- Developed a hybrid framework that combines supervised learning with population-aware anomaly detection to improve BRCA1/2 variant interpretation in data-limited populations.

**Screening Autism Spectrum Disorder in Children Using Speech Transcripts**

Scientific Reports (Nature Portfolio)

R. ASSAF, Z. SHEHABEDDINE, V. RAMESH2025

- Developed speech transcript ML pipelines for ASD screening, offering a privacy-preserving alternative to audio/video modalities.

**A Lightweight Neural Network for Arabic Sign Language Recognition Using Mediapipe Landmarks**

Preprint

Z. SHEHABEDDINE, R. FIDAWI, R. ASSAF2024

- Designed lightweight landmark-based models for static and dynamic ArSL recognition with competitive accuracy and minimal parameters.

# Honors & Awards

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|------|-----------------------------------------------------------------------------------------------------|-----------------|
| 2024 | <b>Dean's Honor List</b> , Recognized for high academic performance                                 | Beirut, Lebanon |
| 2023 | <b>Dean's Honor List</b> , Recognized for high academic performance                                 | Beirut, Lebanon |
| 2022 | <b>Dean's Honor List</b> , Recognized for high academic performance                                 | Beirut, Lebanon |
| 2022 | <b>Late Henri Qais Naccache Competitive Programming Award</b> , Competitive programming competition | Beirut, Lebanon |
| 2021 | <b>ABLE Hackathon Finalist</b> , AI for Accessibility Hackathon                                     | Beirut, Lebanon |